# Personal Knowledge-Base Test Document

***Semantic Retrieval & MCP Testing Notes***

## 1. Semantic Search

Semantic search retrieves information based on meaning rather than exact keyword matches. A user may search for a concept using words that never appear verbatim in the original document. For example, a query such as “finding related information by meaning” should be able to retrieve a passage discussing semantic search even when the exact phrase is absent.

## 2. Embeddings

An embedding model converts text into a numerical vector that represents the semantic properties of that text. Similar pieces of content tend to have vectors that are close together according to a chosen distance metric. In this project, document chunks and user queries are converted into embeddings before being compared in the vector database.

## 3. Vector Database

Qdrant is used as the vector database for this knowledge-base system. Each stored point contains an embedding together with metadata such as the document identifier, user identifier, source name, page number, and chunk index. This metadata allows the retrieval layer to return useful citations alongside the retrieved text.

| Field | Example |
|---|---|
| Collection | knowledge_chunks |
| Distance | Cosine similarity |
| Top-K | 5 |
| Metadata | document_id, user_id, page |

# Retrieval and MCP Testing Scenarios

## 4. Retrieval Pipeline

The retrieval pipeline begins when a user submits a natural-language query. The query is embedded using the same embedding model used during ingestion. Qdrant then searches the stored vectors and returns the highest-ranked chunks. A similarity threshold is applied after retrieval so that the system can reject weak matches instead of returning misleading information.

## 5. Confidence Threshold

A confidence threshold is useful when the knowledge base does not contain information relevant to the user's question. If every retrieved result falls below the configured threshold, the system should return a clear “no confident match” result. The threshold should not be selected arbitrarily; it should be evaluated using a hand-labeled query set.

## 6. MCP Tools

The Model Context Protocol layer exposes the retrieval system as reusable tools. The primary search_notes tool accepts a query and a top_k value and returns ranked chunks with source citations. The get_document tool retrieves the full source context for a document identifier. The list_sources tool enumerates the documents currently available in the user's knowledge base.

## 7. Test Questions

1. What is semantic search?

2. Why are embeddings used in this system?

3. Which vector database is used?

4. What metadata is stored with a vector?

5. Why is a similarity threshold needed?

6. What should happen when no result passes the threshold?

7. What does the search_notes MCP tool return?

8. What is the purpose of get_document?

9. What does list_sources provide?

10. Why should the same embedding model be used for documents and queries?

Suggested out-of-domain test: “What is the capital of Australia?” This question is intentionally unrelated to the document and can be used to test whether the confidence threshold correctly produces no confident match.